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A Methodology for Zumbo's Third Generation DIF Analyses and the Ecology of Item Responding

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Methods for detecting differential item functioning (DIF) and item bias are typically used in the process of item analysis when developing new measures; adapting existing measures for different populations, languages, or cultures; or more generally validating test score inferences. In 2007 in *Language Assessment Quarterly*, Zumbo introduced the concept of Third Generation DIF. In the current article we introduce a new methodology, latent class logistic regression, for Zumbo's Third Generation DIF, whose foundation is a novel ecological model of item responding. The ecological model and the new statistical methodology are introduced, and a proof-of-concept is provided, in the context of an example of an international reading test focusing on DIF due to testing language. The new DIF framework is described and contrasted with other methods, *Mplus* code is provided, and the new method is shown to have potential for application in assessment.

Methods for detecting differential item functioning (DIF) and item bias are typically used in the process of item analysis when developing new measures, adapting existing measures for use in new contexts (i.e., for different populations, languages, or cultures), or more generally validating test score inferences. The purposes of this article are to build on Zumbo's (2007a) description of *Third Generation DIF* by introducing a novel ecological model of item responding and to also

introduce a new psychometric DIF methodology that is particularly well suited for this contemporary view of DIF. It will be shown below in the context of the demonstration that the ecological model and the DIF methodology shine a light on a type of DIF (ecologically moderated DIF) that is obscure or entirely absent from the received standard view of item responding, and DIF in particular. In short, this article aims to provide a proof-of-concept of the novel ecological model of item responding and the use of the new psychometric DIF method. We gratefully acknowledge that this program of research has been influenced by the works of Fox (2001; 2003; 2004), McNamara (1997; 2007), and Chalhoub-Deville (2003).

THIRD GENERATION DIF

In 2007 in *Language Assessment Quarterly*, Zumbo described three generations of DIF; the first generation focused on the motivation of the psychometric problem in test bias and some important concept formation. The transition to the second generation of DIF was signalled by the widespread acceptance of the term “DIF” rather than “item bias,” and the concomitant separation of “impact” and “bias,” and the introduction of new statistical DIF methods that embodied these ideas and distinctions. A few more highlights of the second generation of DIF include (a) a disquieting sense that building the best mousetrap (statistical method) to capture DIF was futile because DIF can occur in so many settings involving varying item formats, item properties (e.g., difficulty, discrimination, and guessing), and sample size configurations that one golden method would not suffice, and hence a full tool kit would be needed (Zumbo, 2007a; 2008) and (b) that flagging DIF, although an important operational task, left us looking for an explanation for why DIF was occurring.

The matter of wanting to know why DIF occurs is an early sign of the third generation of DIF. The multidimensional approach to DIF that grew out of the second generation, in its praxis, places the source of DIF found in a particular group as conceptualized as solely psychological, cognitive, or an unexplained item-specific component in that group. Thus, the multidimensional approach to DIF, while it represents an improvement over psychometric tools used in the first generation, is not a resolution to issues arising in second or third generation DIF.

As Zumbo (2007a) states, *Third Generation DIF* is best characterized by a subtle but extremely important change in how we think of DIF. That is, in the third generation, DIF is conceived as occurring because of some characteristic of the test item and/or testing situation that is not relevant to the underlying ability of interest and hence the test purpose. By adding and highlighting “testing situation” as a possible reason for DIF, one greatly expands DIF praxis and theorizing to matters beyond the test structure itself, hence moving beyond the multidimensional model of DIF that emerged from the second generation.

The shift to a focus on “testing situation” is evident in the work of Fox (2001; 2003; 2004). Fox’s (2003) application of an “ecological approach” (Fox, 2003, p. 21) to bias detection is informed by Bronfenbrenner’s (1979; 1994) considerations of the *environment* and his view that research on behaviour and development needed to extend “beyond the behaviour of individuals to encompass functional systems both within and between settings” (Bronfenbrenner, 1979, pp. 4–5). Fox (2003) appropriates his unit of analysis, the unit of *proximal process* (i.e.,

[person + process + context + time]) (Bronfenbrenner, 1994) in her investigation of “the proximal relationship of raters and test takers, formed during the trialling of a new science version of a high stakes test” (p. 21). She reveals a source of test bias that arises in the interplay between raters, test takers, and test tasks that might otherwise have remained undetected by traditional methods of bias detection, such as those that were common to DIF studies of the day. As Fox (2003) points out, “From an ecological perspective, individuals do not exist as isolated units; rather they are dynamic, socially embedded, and defined by a network of relationships—perceived or actual—occurring in time” (p. 22). Her study highlights factors arising within the testing situation itself that are potential underlying sources of DIF. Throughout her work (see also Fox, 2004; Artemeva & Fox, 2010), the ecology of the testing situation is key in considerations of DIF.

McNamara’s (1997; 2007) research also provides substantive theoretical background to support an ecological approach to DIF. For example, in his 1997 article on interaction in second language performance assessment, he notes that the testing process has a “social dimension” (p. 458), which involves the “interplay of socially derived understandings of the nature and purpose of [testing] activity on the part of the test developers, interlocutors, and raters” (p. 458). A decade later, he (McNamara, 2007) argues that the “greatest challenge facing language testing is the issue of context in which language testing is carried out” (p. 131). He highlights the failure to adequately theorize *context* within language testing research, pointing out that “in standard works on language testing, context has been theorized in terms of the demands it makes on individual cognitive attributes, but this has distorted the picture of the social context available to us” (McNamara, 2007, p. 131). McNamara and Roever (2006) further problematize and explore the social dimension of language testing, lamenting that many in the field continue to characterize the testing situation or social context of testing “in terms of a model of individual ability [which] severely constrains the conceptualization of the assessment context” (p. 32). Others, they argue, have viewed the introduction of the social in considerations of language testing as “an unmanageable aspect . . . opening a Pandora’s box of issues with no chance of practical resolution” (pp. 31–32). McNamara and Roever point out that “DIF detection only considers the test itself and does not look beyond the test booklet at the larger sociopolitical context” (p. 128). They argue that researchers need to go “beyond the test booklet to understand fairness and bias in its social context” (p. 128). They note that “DIF analyses only detect bias, but the factors that cause advantages and disadvantages for groups of test takers and harm their educational opportunities lie in the larger social context” (p. 128).

Bringing to bear the “testing situation,” as Zumbo does in his *Third Generation DIF*, raises the question of the conceptualization of the construct itself. Chalhoub-Deville’s (2003) description of the construct, without loss of generality in her case for language assessment, as an “ability—in language user—in context” resonates with the *Third Generation DIF*. She goes on to state that this view of the construct “claims that the ability components the language user brings to the situation or context interact with situational facets to change those facets as well as to be changed by them” (p. 372). Quite importantly for our approach, she goes on to state that “[t]his perspective, in essence, maintains that ability and context features are intricately connected and it is difficult or impossible to disentangle them” (p. 372).

A NOVEL ECOLOGICAL MODEL OF ITEM RESPONDING

In short, *Third Generation DIF* is part of building an ecological model of item responding and assessment. The ecology of item responding, as Zumbo and Gelin (2005) note, allows the researcher to focus on sociological, structural, community, and contextual variables, as well as psychological and cognitive factors, as explanatory sources of item responding and hence of DIF (Zumbo & Gelin, 2005). For example, a classical example of DIF studies includes a focus on gender-related DIF. However, gender has, in the main, been characterized in the binary as biological sex wherein (binary) biological sex differences on item performance that are eventually explained by item characteristics such as item format and item content. In *Third Generation DIF* “gender” more properly should be considered a social construction, and gender differences on item performance are explained by contextual or situational variables (ecological variables, if you wish), such as institutionalized gender roles, classroom size, socioeconomic status, teaching practices, and parental styles. We believe that these richer ecological variables have been largely ignored in relation to explanations for (and causes of) DIF because of the focus on test format, content, cognitive processes, and test dimensionality that is pervasive in the second generation of DIF.

Building on Zumbo’s (2007a) five general uses of DIF that embody the third generation praxis of DIF analyses, we propose an ecology of item responding as portrayed in Figure 1. Figure 1 depicts the ecology of item responding as embedded in five concentric ovals representing (a) test format, item content, and psychometric dimensionality; (b) person characteristics and typical

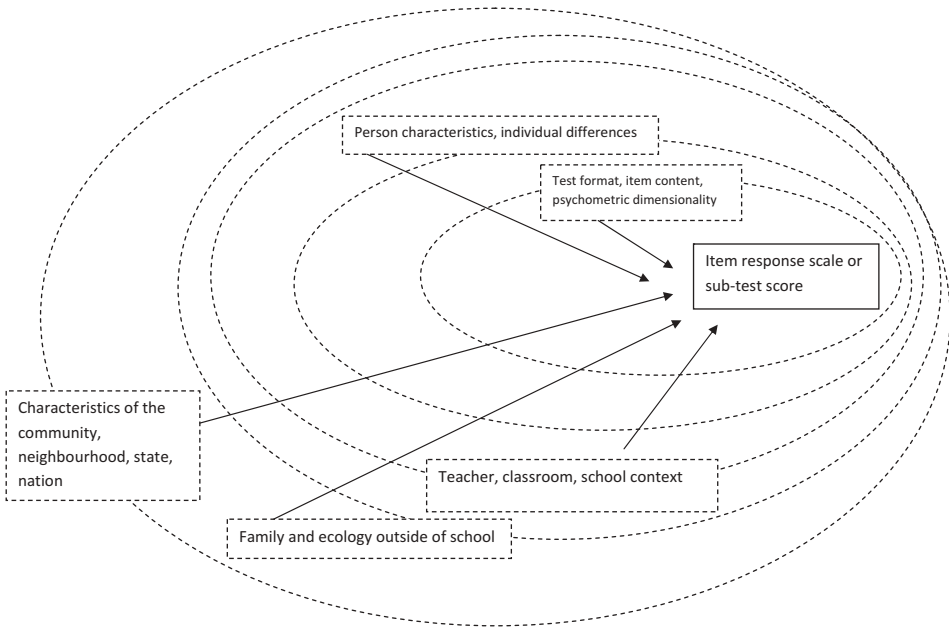


FIGURE 1 An ecological model for item responding.

individual differences variables such as cognition; (c) teacher, classroom, and school context; (d) the family and ecology outside of the school; and finally (e) characteristics of the community, neighbourhood, state, and nation. Conventional first and second generation DIF practices have focused on the first oval with some modest attempts at the second oval as sources for explanation for DIF. It should be noted, however, that we are not suggesting that these are the only five layers of explanatory concepts or variables, nor are we suggesting that they are necessarily measured or manifest explanatory variables. There could be more layers, and these layers could include latent variables in a DIF statistical model, and they could, at least in theory, interact and result in mediated or moderated DIF (Zumbo & Gelin, 2005).

Four points are worthy of note as we transition to the psychometric statistical methods. First, clearly, our model is influenced by ecological systems theory (e.g., Bronfenbrenner, 1979). Second, a natural question arises of how the ecological model's contextual factors could affect item responses by mediating the cognitive processes normally assumed to generate item responses. In response to this question, it is important to keep in mind that we are adhering to the view that neither the test taker nor the cognitive processes in item responding are isolated in a vacuum. Instead, test takers bring their social and cultural present and history to test taking. We accept as our starting point the widely received view in the broader social sciences that human beings have evolved to acquire culture from birth and that the culture to which an individual is exposed, and the ecology of their lives, affects their basic psychology and cognition, including, in our case, item responding. In so doing one can move to a contextualized form of explanation that works against a binary structure of variables that explain test performance. Third, linking DIF to the broader issue of measurement validity, the ecological model further articulates what is meant by "context" in Zumbo's (2009) view of validity as contextualized and pragmatic explanation—that is, the multilayered ecology is the context. Finally, the ecological model serves as a foundation for the statistical and psychometric methodology of DIF analysis.

THIRD GENERATION DIF METHODOLOGY

As an overview of the statistical developments in DIF analysis, DIF methods allow one to judge whether items (and ultimately the test they constitute) are functioning in the same manner in various groups of examinees. In broad terms, this is a matter of measurement invariance; that is, is the test performing in the same manner for each group of examinees? DIF is the statistical term that is used to simply describe the situation in which persons from one group answered an item correctly more often than equally knowledgeable persons from another group. In the noncognitive domain we are not discussing "knowledge," per se but different endorsement (or rating) after matching on the characteristic or attitude or interest. The key feature of DIF detection is that one is investigating differences in item scores after statistically matching the groups on the test score; alternatively, one can condition on the latent ability (see Zumbo, 2007a for a detailed description in which "conditioning" on a variable is akin to treating it as a covariate in a regression analysis). In the conventional cases the "manifest (observed) groups" may represent different groups that you want to compare on the measure of interest, or groups legislated for investigation for bias.

It is useful at this point to describe four frameworks for thinking about DIF that have evolved in the literature: (a) item response modeling via contingency tables and/or regression models, (b) item response theory (IRT), (c) multidimensional models, and (d) latent class methods. Several

essential differences are important to highlight in contrasting the various DIF frameworks. The essential difference between the first two frameworks is that, as Zumbo (2007a) notes, the IRT framework in essence focuses on determining the area between the item response functions or, equivalently, comparing the IRT item parameters of the two groups. Unlike the first framework, the IRT approach does not match the groups by conditioning on the total score. As Zumbo notes, comparing the IRT parameter estimates or item response functions is an unconditional analysis because it implicitly assumes that the ability distribution has been “integrated out”; for example, computing the area between the item response functions across the distribution of the latent continuum of variation, theta. The question of “matching,” as opposed to conditioning in the IRT methods, only comes up if one computes the difference function between the groups conditionally, akin to an analysis of covariance (ANCOVA), as in the Mantel-Haenszel (MH) or logistic regression DIF approaches in the first framework.

The essential difference between the first two frameworks and the multidimensional approach (the third framework) is that the source of DIF in the latter is the psychometric dimensionality of the items, whereas the former focus on the differences in item characteristics such as difficulty, discrimination, and/or guessing.

Contrasting the fourth to the other three frameworks, and particularly the first two, is useful. In the conventional use as practiced in the first two frameworks, DIF methods focus on manifest grouping variables, such as gender, language of the assessment, or ethnicity, and are primarily meant to be used for detecting or flagging potentially problematic items (Zumbo, 2007a). The fourth framework, as highlighted in *Third Generation DIF* methodology, makes use of latent variable mixture models, particularly mixture item response theory (IRT) and mixture Rasch methods for detecting latent groups and testing postulated explanatory models for potential causes of DIF (e.g., Cohen & Bolt, 2005; De Ayala, Kim, Stapleton, & Dayton, 2002; Langeheine & Rost, 1988; von Davier & Yamamoto 2007). The latent class approaches address the question of whether there is a potentially important grouping variable that was not directly observed. The grouping variable need not be a simple manifestation of one variable but may reflect a complex combination of variables that identify important subpopulations (also called latent classes) of test takers. As Sawatzky, Ratner, Kopec, and Zumbo (2012) note, these IRT models specify latent/unobserved classes (in addition to the continuous latent variable, the “theta,” of interest that determines the item responses) that divide a heterogeneous sample into two or more relatively homogeneous groups (also called classes).

Two points are noteworthy here. First, these groups or classes are often referred to as “latent” classes because they are not manifest in the data but rather detected and specified by the statistical model *after* the data and item responses are obtained; as such, the latent classes are estimated simultaneously while the item responses are being analyzed. Second, unlike the conventional continuous latent variables involved in IRT, these latent classes are indicated by discrete latent variables. As Sawatzky and his colleagues go on to note, these mixture IRT models do not produce a general test or measure of DIF on each of the items but rather allow for any of the parameters in the measurement model to vary, or not, across two or more latent classes. As such, one can think of these models as extensions of IRT DIF methods.

The present article introduces a family of latent class logistic regression models that, unlike the previous IRT-based approaches, are extensions of widely used logistic regression DIF methods to allow for latent classes. Traditional models used in DIF logistic regression analysis contain parameters that describe only relationships between observed variables; however, latent class

models (also known as finite mixture models) differ from these by including one or more discrete latent variables. This family of logistic models includes binary, ordinal, and nominal logistic regression and therefore can deal with any of these item response types, or their combination, in an assessment.

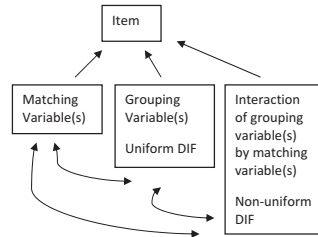
Briefly, to focus on the essential difference of this new DIF method compared to conventional logistic regression DIF, once class enumeration is complete, predictors of class membership allow one to formulate or test explanatory theories of DIF. Figure 2 shows the path models for the four steps in the latent class logistic regression modeling. Step 1 is a conventional logistic regression DIF model depicted in Figure 2 as a path analysis. This first step in the process is optional and simply functions as a comparative reference to the latent class solutions in the remaining steps; that is, one could start at step 2 but then they do not have a comparison to the conventional (one class) logistic regression DIF. It is both customary and useful to depict the model in path analytic form (i.e., manifest variables are squares, latent variables are ovals, single headed arrows are paths or coefficients, and curved double-headed arrows represent correlations). Step 2 is the path model used in the class enumeration. From step 2 in Figure 2, one can see that (a) there is a direct path from the latent class to the item response; (b) dashed arrows lead from the latent class to each of the matching, uniform DIF, and nonuniform DIF, and these dashed arrows are, in essence implied paths in the model that denote the moderated DIF effects. We call this moderated DIF because the dashed lines allow the slopes of the logistic DIF regression to vary across the latent classes; hence, the DIF effects may vary for different latent classes. Step 3 in Figure 2 depicts one manifest variable predicting the latent class, whereas step 4 has multiple predictors.

To summarize the modeling process, the model in step 1 is optional and is used to investigate whether there is DIF using the conventional logistic regression framework, the model in step 2 is used to enumerate (decide upon) the number of latent classes, and the model in step 3 is part of the model-building process to determine which of a list of potential latent class predictors will be investigated together in step 4. Step 2 represents the (latent class) moderated DIF, whereas steps 3 and 4 allow one to investigate the effect of variables in the ecology of item responding (see Figure 1) that may help one understand the complex (moderated) functioning of DIF (Zumbo & Gelin, 2005) by including predictors of latent class membership.

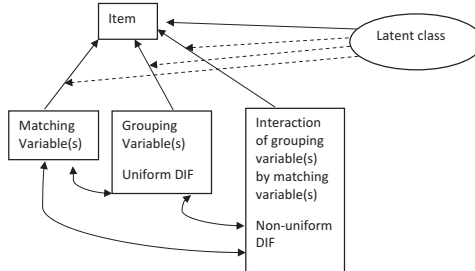
Please note that we are following the most up-to-date latent class modeling procedure recommended by Asparouhov and Muthén (2014). As a brief summary of this new procedure, there are three stages in their computational algorithm. Once the number of classes is decided upon, the next stage involves assigning a most likely class membership for each examinee based on the posterior probabilities obtained in the class enumeration, and the model classification uncertainty/measurement error is computed. In the final computational stage the model uncertainty is fixed on the basis of the information obtained from the prior stage, and the most likely class membership is used as a latent class indicator variable. Next, predictors of class membership can be included in the model noting, of course, that the class membership from the class enumeration is fixed, as described above, and hence does not change once predictors are added.

Let us describe the psychometric theory in the context of an example. We used 2009 data from the Program for International Student Assessment (PISA), an international assessment that measures the performance of 15 year olds in reading literacy, mathematics literacy, and science literacy every 3 years. For our example we investigated the 27 items of booklet number 1 of the reading test and focused on potential DIF based on the test language, French and English versions of the Canadian sample. There were 1,344 students who took the English version and 418 students

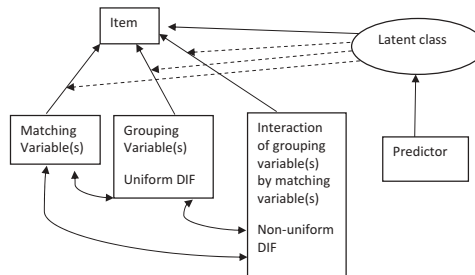
Step #1: Conventional logistic regression DIF (Optional for comparative purposes; go to next step)



Step #2: Model used for class enumeration



Step #3: One latent class predictor at a time.



Step #4: Multiple, denoted by k, latent class predictors

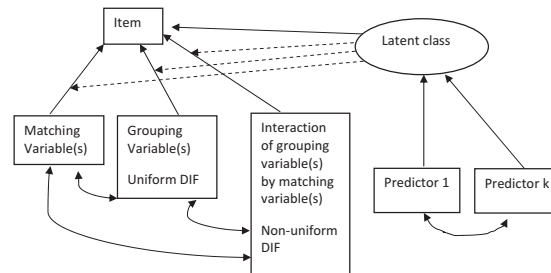


FIGURE 2 The steps in the latent class logistic regression DIF method and the path diagram of the models.

Note. Solid line arrows are direct paths, curved double-headed arrows are covariances, and the dashed arrows from the latent class to each of the matching, uniform DIF, and nonuniform DIF are implied paths in the model that denote the moderated DIF effects. Step 3 depicts one manifest variable predicting the latent class, whereas step 4 has multiple predictors.

who took the French version for a total of 1,762 students. All analyses were conducted by using the Mplus (Muthén & Muthén, 2010) software version 7.11; the Appendix lists the code for the various analyses. Mplus code is provided for steps 1–3, whereas step 4 is simply the third step with more latent class predictors.

The example is solely meant to motivate and give flesh to the abstract statistical concepts in the latent class logistic regression DIF method with an eye toward establishing a proof-of-concept. Our focus is on the statistical method and ecological framework with the purpose of identifying major obstacles and challenges in the implementation of the latent class logistic regression method and the application of the ecological framework.

Step 1: Following step 1 of Figure 2, for each of the 25 binary items we fit binary logistic regression models, whereas for each of the two polytomous items (3-point item scores) we fit the polytomous ordinal logistic regression. A constructed response item (P2000 AESOP Q04; completed by 1,750 students) showed DIF for the language the test was taken in. Both DIF variables (uniform and nonuniform) were statistically significant favouring students taking the English version. In the raw item statistics, of the students who took the English version, 33.3% got it incorrect, 29.7% got a partial mark, and 37% got full points; of the students who took the French version, 37.8% got it incorrect, 33.7% got partial points, and 28.5% got full points.

Step 2: Vermunt (2010) provides a thorough discussion of two longstanding problems in latent class models: the methods used for class enumeration and the modeling strategies when one has predictors of class membership. Following guidance in the statistical literature by Vermunt and others, two statistical tests were used to determine the number of latent classes. In the two-class solution, the Vuong-Lo-Mendell-Rubin likelihood ratio test for one (H_0) vs. two classes (H_0 Loglikelihood Value: -1727.818; 2 times the Loglikelihood difference: 66.538; Difference in the Number of Parameters: 6; p -Value < 0.001) and the Lo-Mendell-Rubin Adjusted likelihood ratio test (Value = 65.085, p -Value < 0.001) were both statistically significant, allowing us to conclude that there are at least two latent classes. When fitting the three-class solution, both the Vuong-Lo-Mendell-Rubin likelihood ratio test for 2 (H_0) vs. 3 classes (p -Value = 0.0840) and the Lo-Mendell-Rubin Adjusted likelihood ratio test (p -Value = 0.0891) were statistically nonsignificant. We also explored the information criteria: the BIC results suggest a two-class solution, whereas the AIC and the sample-size adjusted BIC suggest three classes. As a whole, these statistical tests and fit statistics point to a two-class solution with 76% in class one and the remainder in class two, based on the most likely class membership. Furthermore, 70.9% of the students who took the English version are in class one (remainder of the students taking the English version are in class two), whereas 92.3% of the students who took the French version are in class one (remainder of the students taking the French version are in class two). At this point we can conclude that there are groups of students for whom DIF was found for the test language (French and English); the conventional DIF is not capturing the whole story for the differences in the language of the test.

Comparing the DIF results for the two classes, both uniform and nonuniform DIF are statistically significant for both classes. However, the uniform DIF coefficients are of opposite signs for classes one and two; for those students in latent class one, the uniform DIF effect is such that it favours students taking the English version, whereas for students in latent class two the language DIF effect favours the students who took the French version. Furthermore, plotting the logistic functions for each class separately (see Zumbo, 2008; Oliveri, Olson, Ercikan, & Zumbo, 2012) shows that (a) for class one students taking the English version did better than their French

version counterparts except at the extremes of the score distribution—hence the nonuniform DIF; (b) for class two the students taking the French version consistently outperformed their counterparts taking the English version, with this difference being smaller at the high end of the score distribution.

Step 3: Step 3 is akin to computing correlations (or regressions with only one predictor) prior to a multiple regression. It is meant to aid in the model building by reducing the list of possible predictors of the latent class. Table 1 is a list of the 19 possible predictors of class membership for which data were available. The predictors are listed as per the explanatory levels in the ecological model in Figure 1. These 19 variables were collected as ancillary supplemental variables by PISA. It is important to note both the statistically significant and the nonsignificant predictors because both provide interesting information in understanding what the latent class represents. As can be seen in Table 1, in this example one explanatory variable in each of the categories in the ecological model were statistically significant predictors.

TABLE 1
Outcomes of Step 3, Latent Moderated Logistic Regression DIF With One Predictor at a Time

<i>Levels of Explanatory Variables in the Ecological Model</i>	<i>Sample size</i>	<i>Statistical Significance of Predictor for the Latent Class Variable</i>
Item Content		
Meta-cognition: student reports of the usefulness of the strategy “summarizing” of a long and rather difficult two page text	1,693	$p = .013$
Meta-cognition: student reports of the usefulness of the strategies such as concentration, quickly read, discuss with others for understanding and memorizing the text	1,695	n.s.
Person characteristics, individual differences		
Gender/Sex	1,750	n.s.
Like read – fiction	1,704	n.s.
Like read – non-fiction books	1,699	$p = .001$
Joy/like reading	1,683	n.s.
Teacher, classroom, school context		
At school – Group work	1,690	n.s.
Time – Language lessons	1,685	n.s.
Time – Other language lessons	1,684	n.s.
Teachers stimulation of reading engagement	1,704	n.s.
Teacher student relations	1,708	n.s.
At school – homework	1,688	$p = .025$
Family and Ecology outside of School		
Index of economic, social and cultural status	1,706	n.s.
Amount of time spent reading for enjoyment	1,701	$p = .019$
Highest parental education in years	1,658	n.s.
Wealth	1,711	n.s.
Highest educational level of parents	1,696	n.s.
Home educational resources	1,711	n.s.
Online reading	1,714	n.s.

Note. The sample sizes differ because there were differing missing values in the predictors; “n.s.” denotes a statistically nonsignificant predictor.

Step 4: Finally, a latent class logistic regression model with two latent classes and four predictors (from step 3) of class membership was fit. As a reminder, this model represents the latent class moderated DIF in which we contrast the versions of the test taken in French and English, and the model includes latent class predictors meant to explain the DIF. The three statistically significant predictors in the multiple regression of latent class prediction are (a) the meta-cognition variable indicating the usefulness of summarizing as a strategy (regression-coefficient = -0.182 , $p = .003$), (b) liking to read nonfiction books (regression-coefficient = 0.119 , $p = .027$), and (c) the “at school – homework” (regression-coefficient 0.124 , $p = .047$). It is interesting to note that the “amount of time spent reading for enjoyment” was not statistically significant in the presence of the other explanatory variables. Because the latent class variable has two classes, that is a dichotomous variable, the interpretation of this latent class logistic regression follows that of the conventional logistic regression. Class one was chosen as the reference group because it was the larger group, containing 76% of students, which could give us a more stable comparison especially when one has more than two classes.

Conclusion: At this point it is important to remind ourselves that in contrast to the conventional DIF analysis in step 1 in which we learned that the test language DIF simply favoured students taking the English version, from step 2 of the latent class logistic regression DIF we learned that there is a moderated test language DIF in which we found two latent classes for which the language DIF effect was in opposite directions. That is, students in latent class one who took the English version did better than their French version counterparts except at the extremes of the score distribution. Meanwhile students in class two taking the French version consistently outperformed their counterparts taking the English version, with this difference being smaller at the high end of the score distribution. The overall results suggest that relative to latent class one the students in latent class two are more likely to enjoy reading nonfiction books and spend more time on their school homework but are likely to have lower performance on their meta-cognition skills of summarizing reading materials. In the explanatory model for item responding, this more subtle moderated DIF (i.e., comparing the opposite DIF effects in class one vs. class two) is associated with item content (i.e., the meta-cognitive variable), a person characteristic (i.e., like reading non-fiction), and a teacher/classroom/school context variable (i.e., school homework). Likewise, it is important to also note the explanatory variables that were not statistically significant predictors for the moderated DIF variables reflecting the family and ecology outside of school, as well as the usual suspects in the DIF literature such as gender.

In steps 3 and 4 of the latent class logistic regression method the data analyst is invoking an abductive method of psychometric analysis allowing an inference to the best explanation (Zumbo, 2007a; 2007b). The approach set forth herein proposes that additional explanatory variables for DIF can be identified through exploratory analysis of the situation (i.e., contexts and circumstances) that are hypothesized to enable a test taker’s performance. As we describe above, such variables may pertain to item content, person characteristics of the test taker such as their gender and whether they like or enjoy reading fiction or nonfiction, teacher, classroom and social context, family, and ecology outside of the classroom, and so on. In so doing there is a rhetorical move from how the environment affects the person to a type of interactivism in which the test taker is situated within these enabling conditions and highlights processes and forms of influence of the context/situation (sometimes referred to as the environment) on the test taker that are obscure or entirely absent from the received standard view of item responding and DIF in particular.

GENERAL DISCUSSION

We introduced an ecological model of item responding that serves as a foundation for *Third Generation DIF*. We also extended previous work on logistic regression DIF methods by introducing the latent class logistic regression modeling framework. This framework can be applied to an assessment or test involving any combination of binary or polytomous items. It should be noted that if one were to have a theory or prediction that they were testing about the predictors of the latent class membership, then one could go directly from step 2, the class enumeration step, to step 4, the prediction of class membership, in our process—as opposed to the more abductive model-building approach in step 3.

At this point, it is instructive to contrast the latent class logistic regression approach (that includes covariates of the class membership) with the multiple matching or multiple (logistic) regression approach from the first framework of modeling item responses via contingency tables and/or regression models. These two frameworks are based on different statistical models and hence do not address the same statistical questions nor come to the same conclusions. In general, conventional multiple logistic regression or latent class logistic regression could be used to model the probability of item responding for a binary or ordinal dependent variable as a function of several predictors, including several grouping or matching variables. Latent class logistic regression, however, includes a categorical latent variable(s) where each category represents a homogeneous subpopulation in the overall population of test takers. In essence, in latent class logistic regression, different regressions are estimated for each subpopulation and classify test takers into latent classes (potential grouping variables that were not directly observed) and simultaneously estimate regression models with covariates (predictors) for these latent classes. By contrast, conventional logistic regression assumes that the same model applies to all test takers in the population. Therefore, an advantage of latent class logistic regression over conventional logistic regression models includes relaxing the traditional assumption that the same model holds for all test takers in the population (i.e., conventional regression methods assume one class), allowing the development of separate regressions to be used to target each subpopulation. It is also important, therefore, to note that multiple logistic regression in essence acts as a multiple matching model wherein one investigates manifest group differences controlling for the multiple matching variables, whereas latent class logistic regression investigates whether meaningful subpopulations of test takers are present in the item response data and whether those subpopulations can be characterized (described) by other observed variables. The two approaches (multiple predictors for one-class models in contrast to latent class models) are vastly and fundamentally mathematically different and will only give the same answer if there is one class (no subpopulations) in the population of test takers. Adding multiple predictors (or covariates) to a one-class conventional logistic regression DIF model will, generally, not mimic the properties of latent class logistic regression DIF.

Without lack of generality, the application of the new method and the new ecological model of item responding to international test data provided proof-of-concept and identified the following major obstacles, challenges, and future directions. It is evident from our example that determining the number of classes is not a straightforward exercise and is akin to determining the number of factors in exploratory factor analysis. Furthermore, the day-to-day practice of latent class analysis is constrained by the fact that one is limited to the ancillary or supplemental variables collected along the way in a large-scale testing program, such as the one in our demonstration with the

international assessment. In the future one could test predictions (i.e., confirmatory theory testing) or design experiments in which one is playful of the possible explanatory variables and hence apply a theory-testing strategy rather than the more abductive modeling strategy demonstrated with the international assessment data. These limitations hold for all of the latent class methods in *Third Generation DIF*, whether it is mixture IRT or latent class logistic regression. Regardless of these practical constraints, the latent class logistic regression method and the novel ecological model of item responding that undergirds it, begin to address the issue of the testing situation or context highlighted in Zumbo's Third Generation DIF with a goal of proposing and testing mechanisms or mediators and moderators of situational or contextual effects. This new way of thinking will accelerate progress in explaining DIF and understanding item responding. Future directions in this program of research include extending the modeling to more complicated latent class structures (e.g., von Davier, 2008) as well as allowing for the nested (multilevel) data structures implied in the ecological model of item responding with community characteristic, classroom, school, and neighborhood ecological variables (e.g., Asparouhov & Muthén, 2008; Vermunt, 2008).

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APPENDIX: MPLUS CODE

Mplus Code for Step 1: Conventional Logistic Regression DIF

```
TITLE: this is an example of a logistic regression analysis for DIF analysis
DATA: FILE IS CAN_example.dat;
FORMAT IS 264.F8;
VARIABLE: NAMES ARE id1 id2 i1-i27 total lang totale langc inter;
USEVARIABLES ARE i2 totale langc inter;
CATEGORICAL ARE i2;
MISSING ARE ALL(99);
ANALYSIS: ESTIMATOR = MLR;
MODEL:
i2 ON totale langc inter;
OUTPUT: TECH1;
```

Mplus Code for Step 2: Class Enumeration

```
TITLE: Mixture logistic regression analysis for DIF analysis (class enumeration)
! Two class example
DATA: FILE IS CAN_background.dat;

VARIABLE: NAMES ARE country id1 id2 i1-i27 total lang totale langc inter cond1-cond22;
USEVARIABLES ARE i2 totale langc inter;
AUXILIARY = cond1-cond22;
```

```

CATEGORICAL ARE i2;
MISSING ARE ALL(99);
CLASSES = c (2);

```

```

ANALYSIS: TYPE = MIXTURE;
STARTS = 0;
LRTSTARTS = 0 0 500 100;

```

```

MODEL:
%OVERALL%
i2 ON totalc langc inter;
%c#2%
i2 ON totalc langc inter;

```

```

SAVEDATA: FILE = step2.dat; SAVE = CPROB;
OUTPUT: TECH10 TECH11 TECH14 CINTERVAL;
PLOT: TYPE = PLOT3;

```

Mplus Code for Steps 3 or 4

```

TITLE: this is an example of a mixture logistic regression analysis for DIF analysis
DATA: FILE IS step2.dat; !use the data saved from step2
FORMAT IS 29F10.3;

```

```

VARIABLE: NAMES ARE i2 totalc langc inter cond1-cond22
p1 p2 N;
USEVARIABLES ARE i2 totalc langc inter cond22 n;
CATEGORICAL ARE i2;
nominal = n;
MISSING = *;
CLASSES = c (2);

```

```

ANALYSIS:
TYPE = MIXTURE;
STARTS = 0;

```

```

MODEL:
%OVERALL%
i2 ON totalc langc inter;
c ON cond22;

```

%c#1%
[N#1@2.299];

%c#2%
[N#1@-0.414];
i2 ON totalc langc inter;

OUTPUT: TECH1 CINTERVAL;
